7.5 Practice: percentages (*procenten*)

← Unit 7.5 · Answers

Remember the friendly percentages: 100% is everything, 50% is half, 25% is a quarter, 10% is a tenth, 1% is a hundredth. You can draw a bar (*strook*) or a ratio table (*verhoudingstabel*) for any question.

Set 1 · Warm-up: what percent means

1. 100% of 40 =
2. 50% of 40 =
3. 50% of € 12 =
4. 25% of 100 =
5. 25% of 80 =
6. 10% of 70 =
7. 10% of 350 =
8. 1% of 400 =
9. On a 10 × 10 grid, 50 squares are coloured in. What percentage is coloured?
10. You eat 100% of a pizza. How much pizza did you eat?

    Talk about it: Questions 6 and 7 are both “10% of”. How do you work out 10% of any number?

Set 2 · Building: percentages and fractions (*procenten en breuken*)

Write as a fraction (as simple as possible):

1. 50% =
2. 25% =
3. 75% =
4. 10% =
5. 20% =

Write as a percentage:

6. 1/5 =
7. 3/10 =
8. 1/100 =
9. 2/5 =
10. 1/3 = about …%
11. 1 whole =

    Talk about it: Why does 1/3 not give a “nice” percentage, but 1/5 does?

Set 3 · Building: building from 50%, 10% and 1%

1. 20% of 60 =
2. 30% of 50 =
3. 5% of 80 =
4. 75% of 40 =
5. 15% of 200 =
6. 40% of 25 =

7. 90% of 50 =
8. 12% of 300 =
9. 60% of 45 =
10. 50% of 10 =

Talk about it: For question 7, did you build up from 10%, or did you take something away from 100%?

Set 4 · Building: discount (*korting*)

For each one, work out how much you save *and* the new price.

1. A T-shirt costs € 20. Korting: 50%.
2. Trainers cost € 60. Korting: 25%.
3. A computer game costs € 40. Korting: 10%.
4. A bike costs € 250. Korting: 20%.
5. A winter jacket costs € 80. Korting: 30%.
6. Ice skates cost € 45. Korting: 10%.
7. Packs of stroopwafels cost € 3 each. Offer: *2e halve prijs* (the second pack is half price).
   a. What do two packs cost?
   b. What percentage do you save on the two packs together?
8. A 400 g pack of muesli has "25% extra gratis". How many grams are in the pack now?
9. A book costs € 10. Korting: 50%. What do you pay?

Talk about it: In question 8, is "25% extra" the same as "25% korting"?

Set 5 · Stretch: word problems and deals

1. In a class of 25 children, 20 come to school by bike. What percentage is that?
2. A quiz has 40 questions. You get 30 right. What percentage is that?
3. Your tablet is at 80%. After a film it's at 35%. How many percent did the film use?
4. Which is cheaper?
   - Shop A: € 30 with 20% korting
   - Shop B: € 28 with 10% korting
5. A child's train ticket has 40% korting on the full price of € 15. What does the child's ticket cost?
6. Of the 200 pupils at a school, 15% walk to school. How many pupils walk?
7. True or false: if something has 50% korting, and then another 50% korting on the new price, it becomes free. Explain.
8. Whole milk contains about 3,5% fat. How many grams of fat are in 1 000 g of milk?
9. 10% of 100 =

Talk about it: In question 4, the bigger percentage gave the cheaper price. Is that always true?

Puzzle corner (optional)

The swap trick. Which is bigger: 10% of 50, or 50% of 10?

- Try a few more pairs: 20% of 5 and 5% of 20. 25% of 8 and 8% of 25.
- What do you notice? Does it always work?
- Use it to work out 8% of 25 in your head the easy way.